American Birds (revisited)

Last week we found a **positive relationship** between your *anchor* (X=50 or X=70) and the value you picked for the percentage of American birds found in California. That was a model of a relationship between a **categorical predictor** (X) and a **continuous response**.

But consider the possible effect of *prior knowledge of birds*. Represented by the score on the bird quiz, prior knowledge is a **continuous predictor** that could have an additive effect on the value chosen for the percentage of American birds found in California.

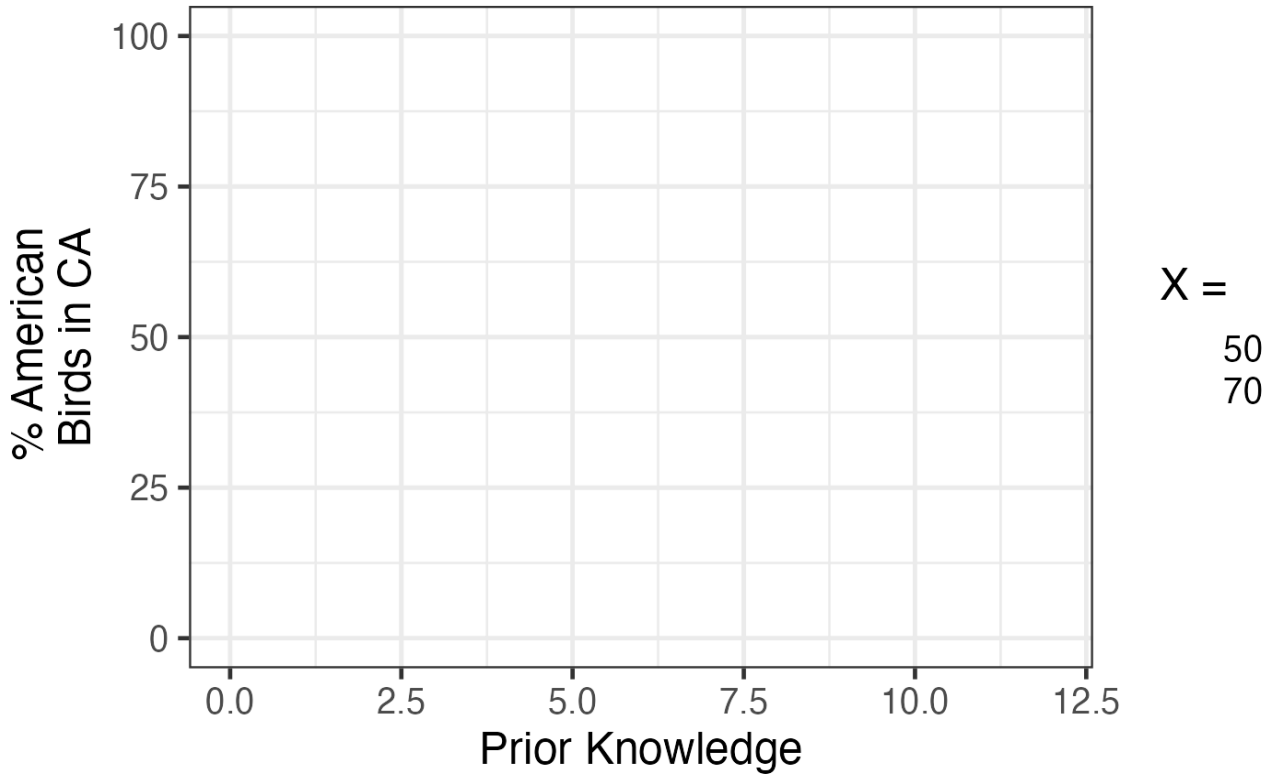
Describe your hypotheses

Write the model for the multiple regression model in statistical notation. You should have a **continuous response**, a **categorical predictor**, and a **continuous predictor**. There should be three coefficients total: one for each predictor plus an intercept.

The data from the classroom experiment are shown on the screen. In your group, discuss what effect you think the two predictors had on the response. Pick approximate values for your coefficient estimates.

$\beta_0 \approx$ $\beta_1 \approx$ $\beta_2 \approx$

On the following page, draw two lines (one solid, one dashed) representing your coefficient estimates. One line should represent observations where X=50, the other line X=70. Fill in the legend to indicate your choice.


100
75
50
25
0
% American Birds in CA
0.0
2.5
5.0
7.5
10.0
12.5
Prior Knowledge
X =
50
70